Control and Mapping Strategies for *HybriD*

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Abstract

HybriD is a collaborative work in progress between contemporary music group re-sound and the author for multiple acoustic instruments and live computer processing and synthesis. This article outlines some of the strategies explored and methods employed in the development of software to explore the creation of "hybrid" sounds based on the combined elements of multiple performers' musical gestures.

1. Introduction

HybriD is a collaborative work in development between contemporary music group *re-sound* and the author, for multiple acoustic instruments (2 saxophones, 2 clarinets, violin, electric bass), MIDI Percussion instrument and live computer processing and synthesis. In many ways, this piece builds on the ideas initially formed during the composition of *Nexus*, which was commissioned by the contemporary performance group *Australysis* in the mid 90s. In that piece, the interpretation of MIDI-based performance data from three different performers was gradually merged to create a flavour of 'Hyperinstrument' (Machover and Chung 1989). With the given advances in computer processing power and software design, new methods of merging and mapping real-time performance data become available and some approaches are explored in this piece. Additionally, the resources offered by a relatively large ensemble offer multi-dimensional control of synthesis and signal processing parameters to achieve an electro-acoustic sound world with the potential of great depth and complexity, much of which is offered by the detail and nuance provided by an instrumental performer.

Development of the software and writing and refinement of the piece has been an ongoing affair. Individual workshops with ensemble performers were held during March and April and workshops with the entire ensemble commenced as of May 2002. At the time of writing, the ensemble has had the opportunity to 'play with the piece' in a few of sessions. These sessions have already provided valuable feedback on the efficacy of various mapping strategies and have served to prompt further development of many aspects of the software.

In its current form, the piece is realized with Max/MSP object-oriented programming software running on a 550MHz Apple Titanium Powerbook. Audio I/O is provided via a Mark of the Unicorn 828. Though a contemporary system, its performance benchmarks are relatively modest and the issue of maximizing CPU efficiency is an ever-present consideration in the design of the software.

2. Audio Analysis and Derived Control Sources.

Given the reliance that this work has on instrumental performance as a means of control, methods of tracking and representing performance parameters through an instruments audio signal are an important factor. Though a detailed investigation into this area is beyond the immediate scope of this project, a brief description of the means employed is appropriate.

The audio signal of each acoustic instrument, captured by instrument microphones or 'bugs', is fed to an analysis stage in the *HybriD* program (a Max/MSP patch) as a sampled audio signal. The MSP object *fiddle~* (Puckette et al.1998) is employed to track real-time pitch and amplitude characteristics of each signal. *fiddle~* is freely available, fairly efficient, and above all, capable of providing generally useful results for the purposes of this work. Each analysis stage generates an output in the form of discrete notes (in MIDI note format) and continuous pitch and amplitude outputs at the Max scheduler (or control) rate. Audio signals from each of the instruments are

also available as sources for routing to real-time signal processing algorithms. Coupled with the MIDI performance data provided by the ensemble's percussionist via pads and data from the computer operator via a fader based MIDI controller and computer keyboard, the outputs of the analysis stages make up the available complement of externally derived control sources.

Drawing on software previously developed with Max for the composition *Nexus*, some feature detection in the form of performance data (MIDI note information, duration and inter-onset time stored in cyclic buffers of arbitrary dimensions) is also undertaken. This data is used to track tempo in a rudimentary manner and generate continuous control sources intended in part to facilitate experimentation with controller mapping.

Finally, some control sources are generated internally, such as Low Frequency Oscillators (LFOs) and breakpoint envelopes. Numerous parameters of these sources can be controlled by the performance control sources described above. For example, the rate and amplitude parameters of an LFO can be modified by the pitch and amplitude control values generated by a performer.

3. Mapping Strategies

Within the context of this work, emphasis is placed on utilizing the resources of the ensemble through the exploration of interactive networks that can be formed between performers and the resultant sound(s) they may produce. The ability to define a path of influence for each performer's input to the system independently is necessary, and consequently, a practical method of coupling performance features (control sources) to the processes that determine the sonic results (synthesis parameters) is one of the first considerations.

Using Max/MSP, the routing of control and audio sources to a variety of destinations is most readily achieved by using a number of switching matrices similar in concept to the patching panels popular in analogue synthesizers. An arbitrary number of states of each matrix can be stored and recalled for later use. In its current guise, *HybriD* uses two matrices, one to route control sources and the other audio.

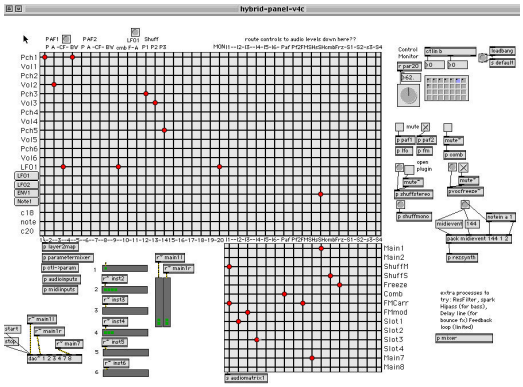


Figure 1. A view of the *HybriD* Matrix Panel showing the larger upper control routing matrix and the lower smaller audio routing matrix.

The two matrices are positioned in the way that they are so that control sources can be routed to effect audio channel parameters, such as volume and panning. Aside from the fact that legible representations of these matrices take up a lot of screen space and become difficult to navigate as larger numbers of control sources and destinations are accommodated, their use is straightforward. Those functions that are less frequently called upon in the development of the software can be assigned to the matrix via a menu when required. This enables a larger number of potential control sources to be available than the dimensions of the matrix can accommodate, albeit not all simultaneously.

The issue of interconnection becomes one of greater significance when considering the suitability of the control source to its intended destination. As an example, *fiddle~* provides pitch data in the form of a floating point MIDI Note value where each equal-tempered semitone is represented by a value of 1.00 (for 100 cents). In this form, the typically encountered pitch range of a violin is conservatively around 36 (3 octaves) commencing from 55 (G below middle C). If this value range were used to control the gain of an audio buffer which would typically operate in floating point values between the range of 0 to 1, we would have a massively distorted signal as a result.

From this example, questions of how a particular pitch range as extracted from a performance should affect the amplitude of a sound (or even why this should be done at all) arise. Removed from the context in which these pitch values are derived and

placed in a context where they have little if any meaning, there can be no single correct response. Nevertheless, the facility to arbitrarily map a source to a destination is an implicit requirement of the structure.

Setting aside the consideration of linear and logarithmic characteristics inherent in various control sources, an initial solution to the problem of managing extremely large (or small) values is to scale all control sources and destinations so they operate within a standardized range. Doing so enables any control source to be coupled to any synthesis or processing parameter with confidence that the results will fall into an acceptable (if somewhat arbitrary) range.

Within the domain of research into performance-based synthesis control there is a general consensus that simple one-to-one mapping of control sources to synthesis parameters rarely yields the most useful or interesting results (Hunt et al. 2000) (Garnett and Goudeseune 1999). (One-to-one mapping is implicitly illustrated in our earlier example where the detected pitch of one instrumental source is used to control the amplitude of a - possibly unrelated - sound source but imposes no other influence in the way the resulting sound behaves.) Comparing the dynamic interplay of sonic qualities of acoustic instruments or, more appropriately, aesthetically appealing synthesis algorithms, to synthesis algorithms that employ only one-to-one mapping quickly bears this observation out.

Enabling the capacity to merge control sources (many-to-one) of variable weights or alternately, having numerous synthesis parameters each affected by differing amounts in response to the same control source (one-to-many) raises numerous questions regarding how such a system might operate and what implications this has for the architecture of the system.

The use of an intermediate mapping layer as proposed elsewhere (Wanderley et al. 1998) proceeds by dividing the mapping scheme into independent layers that more readily enable the expressive use of different input devices (acoustic instruments or MIDI in this case) to control a variety synthesis or processing algorithms. This proposal serves as a model for implementation within HybriD.

A model for the interconnection of control sources to parameter destinations grows in complexity and inefficiency as new, combinatorial possibilities, transformative processes and special cases are added. On the other hand, a model that can deal with the most frequently required mappings enables it to be used often and the occasion of conditions for which unique treatments need to be devised are encountered less frequently if at all.

As it currently stands, the scheme is fairly primitive, however one-to-one, one- to-many, many-to-one and many-to-many controller mappings are, to a limited extent, available. This is achieved by way of routing sources to a secondary mapping layer or control mixer. The number of control sources in any control mixer is currently limited to four, though an additional mixer can be used to increase the number of control sources that can be combined if they are required. A similar arrangement is available for distribution of one control source to multiple synthesis parameters.

One of a number of problems that arise with the current implementation is that the control mixing connections made are not reflected in the visual state of the connection matrix. In order to maintain consistency with this representation, a more elaborate system of routing control sources is planned, but not yet implemented. This would most probably require the ‘tagging’ of control sources so that they could be merged into a single data stream (i.e. the matrix buss to which they are assigned) and identified by their tag at the receiving end.

There are a number of possibilities by which two or more different control sources may be combined. Varied weighting between different control sources by setting a ‘tap point’ along a linear or logarithmic curve between two or more states similar to the way in which a panning control in an audio mixer operates is one method. This is equivalent to the addition of two weighted values. The use of one control source to determine the range of another by multiplication is also possible. Both methods are currently in use in HyBriD, however further refinement and generalization of the treatment and combination of control sources is left as an area for future work.

4. Signal Processing network

The audio system provides initial sources in the form of the captured instrument audio signals and a number of interchangeable synthesis algorithms. Signal processing elements, busses and output

channels, can accept inputs from all sources (except their own outputs in most cases) via the matrix. Initially, the algorithms used were all rather complex but after some experimentation, it has been found that using a larger complement of simpler processing elements enables finer structural control of the algorithms, better integration between the control sources to the sound production process and reduces the need for interpolation of multiple parameters simultaneously (as is the case with more complex algorithms). The only real disadvantage of this approach is the requirement for a greater number of interconnections in the audio matrix arises which leads to inefficiency and conceptual difficulties.

Some of the synthesis algorithms used include simple oscillators, additive (oscillator bank) and formant synthesis. Signal processing algorithms in use include various types of filtering including filter banks, frequency modulation, amplitude modulation and spatialization, delay lines and comb filters, granulation and looping among others. As with the control matrix, some of the less frequently used elements are relegated to channels from which they can be selected from a pool of processes and enabled when a signal is routed to them. This serves to keep the audio matrix to a manageable size and reduce processor overhead by switching off unused components that would otherwise consume valuable CPU resources.

The process of determining successful mappings in this work is largely as the result of experimentation during workshop sessions with the ensemble or, more frequently at this point in time, by simulation of a performance via playback of pre-recorded fragments of the performers in action.

While the potential exists to map a large range of sources to destinations, resulting in a huge range of permutations, it is likely that only a relatively small subset of all possible mappings will be effective for the stated aims of the work. The general pattern observed is that better results are achieved by maintaining a balanced distribution of the influence of the performers on the synthesis process. This balance becomes more difficult to manage as the number of contributing performers increases toward the full complement of the ensemble.

To illustrate the kinds of mappings possible, a practical example is provided: A cluster of sine tones is generated using an additive synthesis algorithm. The pitches of two instruments can be used to independently determine the lower and upper boundaries of the oscillator frequencies while the overall volume of the cluster may be controlled by a third player. A fourth performer's input could be used to influence spatial disposition of the resultant sound across a multi-channel sound system. In this example, many performers are involved and simple one-to-one mapping of control source to synthesis parameter is predominant. By reducing the number of performers (and consequently, control sources) the scenario changes: If the pitch of one performer were used to control the lower frequency boundary, while a function of pitch and amplitude (pitch, scaled by amplitude) from the same performer were used to determine the upper frequency boundary and yet another variant of these two control sources (a weighted sum of amplitude and pitch) was assigned to control the volume of the resultant sound, both one-to-many and many-to-one mappings arise.

Aural results that reflect a high degree of expressive reaction to the performers' sonic gestures are the primary goal, though there is no real reason to reject mappings that don't strictly fulfill this requirement. In a few cases at least, subtle processing of instrumental audio signals makes up the entire output of the system. Processing parameters may not react in any way to the performers' gestures but simply process the audio signals in a fixed manner over a given period of time.

It is often the case that a particular mapping yields promising aural results from a composer's (or listener's) perspective, while the performers' sense of their affect on the resulting sound may be counter-intuitive or even difficult to determine and thus prove less engaging for them. As the mapping is likely to mutate frequently in a developed performance of the work, each performer's relationship to the sound will change with each mutation, offering performers little opportunity to get comfortable with any particular mapping.

5. Future Work

Ongoing development of the *Hybrid* software offers new possibilities and refinements, but doing so between rehearsal sessions with the ensemble does have negative implications for development of the work overall. It is expected that the further development of the software will be suspended in the

near future so that a greater sense of stability ensues and mappings that yield results that are consistent from one rehearsal to another are possible. It should be pointed out however, that even when mappings are totally reproducible from one session to another, there is no guarantee (or expectation) that the aural results will be identical due to variations in performance, bug placement and any number of other variables.

At this time, no tracking or estimation of the timbral or spectral qualities of the captured sounds is used as control sources. This limitation is in part the result of CPU efficiency considerations as tracking and processing six instruments in real-time (among other DSP tasks) places fairly heavy burdens on the processor. Future work with a single performer or a small group will provide greater scope for exploration in this area.

In moving between different mappings the structure of the program actually changes at both the control and audio level. While gradual alteration of states in the audio matrix is possible, control mappings are switched from one state to another more or less instantaneously. It is not anticipated that this limitation will be easily resolved, though possible methods of interpolation between different states need to be investigated.

In its current form, complex mapping is a tedious affair. It is anticipated that further development of practical and effective mapping strategies of performance-control sources to synthesis parameters as outlined in this article will be enhanced by the experience gained from the work on this project.

6. Acknowledgments

Thanks to Dr Thomas Reiner and the performers of *re-sound* for their generous support in the realization of this work. This project has been assisted by the Commonwealth Government through the Australia Council, its arts funding and advisory body.

References

Garnett, G. E. and C. Goudeseune. 1999. “Performance Factors in Control of High-Dimensional Spaces.” *In Proceedings of the 1999 International Computer Music Conference*, San Francisco: International Computer Music Association. pp. 268-271

Hunt, A., M. M. Wanderly and R. Kirk. 2000. “Towards a Model for Instrumental Mapping in Expert Musical Interaction.” In *Proceedings of the International Computer Music Conference*, San Francisco: ICMA.

Machover, T., and J. Chung. 1989. “Hyperinstruments: Musically Intelligent and Interactive Performance and Creativity Systems.” In *Proceedings of the 1998 International Computer Music Conference*. San Francisco: ICMA

Puckette, M. S., T. Apel, D. Zicarreli. 1998. “Real-Time audio analysis tools for Pd and MSP.” In *Proceedings of the 1998 International Computer Music Conference*. San Francisco: ICMA. pp. 109-112

Rowe, R. 1993. *Interactive Music Systems*. Cambridge, Mass: MIT Press.

Wanderley, M. M., N. Schnell, et al. (1998). “ESCHER - Modelling and Performing Composed Instruments in real-time.” In *Proceedings of the 1998 IEEE International Conference on Systems, Man and Cybernetics* (SMC). Pp 1080-1084.

Weinberg, G. and S. L. Gan. 2001. "The Squeezables: Toward an Expressive and Interdependent Multi-player Musical Instrument." *Computer Music Journal* 25(2): 37-45.